MIL-P-51062B
5 June 1969
SUPERSEDING
MIL-P-0051062A(MU)
31 March 1966

MIL-P-51062
5 December 1961

MILITARY SPECIFICATION

PUMP, FOG OIL, SMOKE GENERATOR, M4

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a fog oil pump for use with the ABC-M3A3 smoke generator.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-W-572 - Wood Preservative; Water Repellent.
PPP-B-585 - Boxes, Wood, Wirebound.
PPP-B-591 - Boxes, Fiberboard, Wood-Cleated.
PPP-B-601 - Boxes, Wood, Cleated-Plywood.
PPP-B-621 - Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636 - Box, Fiberboard.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes.

FSC 1040

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MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-T-704 - Treatment and Painting of Materiel.
MIL-F-12070 - Fog Oil.
MIL-C-17504 - Coating Compound, Acrylic, Clear.
MIL-L-21260 - Lubrication Oil, Internal Combustion Engine Preservative and Break-in.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

DL31-15-1197 - Pump, Fog Oil, Smoke Generator, M4 - Bill of Material.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Ratings, Rules and Regulations.

(Application for copies of these freight classification rules should be addressed to the Uniform Classification Committee, 202 Union Station, Chicago, Illinois 60606.)

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3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL31-15-1197 or on subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

3.1.2 Components. All components of the pump shall conform to the specifications and drawings listed on Drawing DL31-15-1197 and subsidiary drawings.

3.2 Assembly. The pump shall be assembled as shown on Drawing E31-15-1197.

3.3 Leakage. The pump assembly shall not leak air when subjected to an internal pressure of 25 ∠ 10 seconds when tested in accordance with 4.4.4.1.

3.4 Functioning. The pump assembly shall deliver a minimum of 55 gallons per hour (gph) of fog oil when operated at a test pressure of 3 ∠ 1/2 psig. The pump shall operate freely and continuously (that is, without stopping or stalling) at test pressure for 15 ∠ 2 minutes, with fog oil being emitted from the ends of both the oil exhaust and oil outlet lines when tested in accordance with 4.4.4.2.

3.5 Endurance. The pump shall operate continuously for 7 ∠ 1 hours at the test pressure specified in 3.4. The rate of delivery as specified in 3.4 shall be determined at 1-hour intervals when tested in accordance with 4.4.4.3.

3.6 Preproduction. Prior to the start of regular production, a preproduction sample of pumps shall be produced for examination and test in accordance with 4.3.

3.7 Workmanship. The pump and components shall be free from damage which could affect its intended use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsibile for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his

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own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3)
(b) Quality conformance inspection (see 4.4)

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of two pumps shall be produced using the same methods, materials, and equipment that will be used in regular production.

4.3.2 Inspection procedure. The pumps shall be tested in accordance with 4.4.4.1, 4.4.4.2 and 4.4.4.3.

4.3.3 Acceptance/rejection criteria. The preproduction pumps shall meet all the tests of this specification to be acceptable. The supplier shall obtain written approval from the contracting officer prior to starting regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the pumps of one interfix number, produced by one manufacturer, at one plant, for the same materials, and under essentially the same manufacturing conditions.

4.4.2 Sampling for examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105. For endurance test general inspection level 1 shall be used.

4.4.3 Inspection procedure.

4.4.3.1 Examination and nondestructive tests. Sample components, pump assemblies and the levels A and B preparation for delivery shall be examined in accordance with the classification of defects and with MIL-STD-105. For endurance test use an AQL of 2.5 percent defective.

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4.4.3.2 Critical characteristics. Each pump in the lot shall be examined for the critical characteristics listed in the classification of defects.

4.4.3.3 Classification of defects.

(a) Bushing (Dwg B31-15-1299).

Categories		Defects
Critical:		None defined
Major:	AQL	1.0 percent defective
101		Inside diameter incorrect
102		Diameter 0.9360 incorrect
103		O-ring groove diameter or width incorrect
104		Thread incorrect
105		Eccentricity excessive
106		Machine finish incorrect
107		Workmanship (3.7)
Minor:	AQL	2.5 percent defective
201		Plating missing or incorrect

(b) Bushing (Dwg B31-15-1199).

Categories		Defects
Critical:		None defined
Major:	AQL	1.0 percent defective
101		Thread incorrect
102		Inside diameter incorrect
103		O-ring groove diameter or width incorrect
104		Machine finish incorrect
105		Workmanship (3.7)
Minor:	AQL	2.5 percent defective
201		Plating missing or incorrect

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4.4.3.3 Classification of defects (continued).

(c) Tappet, valve (Dwg B31-15-1218).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Diameter (0.311) incorrect
102	Machine finishes incorrect
103	Overall length incorrect
104	Hardness incorrect
105	Flatness beyond tolerance
106	Workmanship (3.7)

(d) Rod, pump assembly (Dwg B31-15-1219).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Diameters (0.499) incorrect
102	Overall length incorrect
103	Eccentricity excessive
104	Machine finish incorrect
105	Hardness incorrect
106	Diameter (0.300) incorrect
107	Length (2.000) incorrect
108	Workmanship (3.7)

(e) End, cylinder (Dwg C31-15-1236).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Mounting holes incorrect or incorrectly located
102	Thread (5/8-18) incorrect
103	Diameter (3.490) incorrect
104	Diameter 0.375 incorrect or incorrectly located
105	Slot width incorrect
106	Workmanship (3.7)

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4.4.3.3 Classification of defects (continued).

(f) Cylinder, air (Dwg D31-15-1239).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Thread (5/8-18) incorrect
102	Thread (1/2-14NPT) incorrect
103	Diameter (0.375) hole incorrect or incorrectly located
104	Slot width (0.380) incorrect
105	Tapped holes (5/16-18) and 1/4-20) incorrect or incorrectly located
106	Diameter (0.402) holes incorrect or incorrectly located
107	Diameter (0.876) incorrect
108	Diameter (0.389) incorrect
109	Width and diameter of "O" ring groove incorrect
110	Diameter (3.502) incorrect
111	Eccentricity excessive
112	Machine finish 32 incorrect
113	Perpendicularity incorrect
114	Thread (1/8-27NPT) incorrect
115	Workmanship (3.7)
Minor:	AQL 2.5 percent defective
201	Plating missing or incorrect

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4.4.3.3 Classification of defects (continued).

(g) Cover, valve (Dwg D31-15-1300).

Categories		Defects
Critical:		None defined
Major:		AQL 1.0 percent defective
101		Thread (5/8-18) incorrect
102		Thread (7/8-14) incorrect
103		Thread (1 1/8-18) incorrect
104		Diameter (0.402) holes incorrect or incorrectly located
105		Eccentricity excessive
106		Parellelism incorrect
107		Machine finish incorrect
108		Depth (1.468) incorrect
109		Tapped holes (1/4-20) incorrect or incorrectly located
110		Diameter (.1250) incorrect or depth (0.188) incorrect
111		Diameter (0.9365) incorrect
112		Perpendicularity incorrect
113		Thread (1/8-27NPT) incorrect
114		Workmanship (3.7)

(h) Valve, yoke operating (Dwg B31-15-1244).

Categories		Defects
Critical:		None defined
Major:		AQL 1.0 percent defective
101		Hole diameter incorrect
102		Diameter (0.310) incorrect
103		Machine finish incorrect
104		Hardness incorrect
105		Centerline of hole to shank shoulder incorrect
106		Workmanship (3.7)
Minor:		AQL 2.5 percent defective
201		Overall length incorrect
202		Plating missing

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4.4.3.3 Classification of defects (continued).

(i) Pump, fog oil, smoke generator, M4 (Dwg E31-15-1197).

Categories	Defects	Acceptance standards
Critical:		
1	Leakage	4.4.4.1
2	Functioning	4.4.4.2
Major:	AQL 1.0 percent defective	
101	Component missing or incorrectly assembled	
102	Marking missing	
Minor:	AQL 2.5 percent defective	
201	Protective finish missing	

(j) Rod, piston (Dwg B31-15-1226).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Shoulder length incorrect
102	Overall length incorrect
103	Eccentricity excessive
104	Machine finish incorrect
105	Threads incorrect
106	Shoulder diameter incorrect
107	Diameters (.249) incorrect
108	Straightness incorrect

(k) Valve body (Dwg C31-15-1237).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Disk pressurizing valve cavity incorrect
102	Disk retainer cavity incorrect
103	Ring groove incorrect
104	External thread incorrect
105	Internal thread incorrect
106	Diameter of "O" ring cavity incorrect
107	Machine finish incorrect
108	Diameter (1.008) incorrect

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4.4.3.3 Classification of defects (continued).

(l) Disk, pressurizing valve (Dwg A31-15-945).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Diameter incorrect
102	Flatness incorrect
103	Workmanship (3.7)

(m) Retainer, disk (Dwg B31-15-1301).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Outside diameter incorrect
102	Flatness incorrect
103	Workmanship (3.7)

(n) Preparation for delivery (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 4.0 percent defective
101	Surfaces not cleaned as specified
102	Openings not covered as specified
103	Internal fiberboard blocking insufficient to make a tight pack
104	Container damage (broken, torn, punctured, or wracked)
105	Marking incorrect, missing, or illegible

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4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Leakage. Plug the air intake and oil outlet ports of the pump assembly. Pressurize the pump assembly to the specified pressure by introducing air through the oil inlet and oil exhaust ports. Submerge the pump assembly in water for the required period of time.

4.4.4.2 Functioning. Mount the pump assembly on a suitable stand and connect 7-to 10-foot lengths of hose or tubing to the oil inlet, oil outlet, and oil exhaust ports. The oil inlet hose shall contain a suitable sediment strainer. The oil outlet hose shall contain a metering orifice with a 0.180 / 0.010-inch diameter by 1/8 / 1/32 inch long hole. Place the hose or tubing lines into a container or type SGF2 fog oil, conforming to MIL-F-12070. Place the end of the inlet line below the surface of the fog oil and the ends of the outlet and exhause lines above the surface of the fog oil. Connect a regulated source of filtered compressed air to the pump assembly air inlet port. Open the pump assembly bleed valve and introduce air into the pump at an initial pressure of 5 / 1 psig. When air exhausts from the bleed valve, close the valve and operate the pump for a break-in period of 15 minutes. If necessary, initial starting or restarting of the stalled pumps, during this break-in period, may be facilitated by manually pushing on the pump air motor rocker arms. Immediately following the break-in period, without stopping the pump, the air pressure shall be reduced to the specified test pressure. Place the ends of the outlet and exhaust lines into a volumetrically calibrated container to determine the delivery rate of the pump.

4.4.4.3 Endurance. Operate the pump in accordance with 4.4.4.2 for the specified time, determining the delivery rate of the pump at the specified intervals.

5. PREPARATION FOR DELIVERY

5.1 Packaging and packing. All packaging and packing shall be of the minimum cube required to contain the items.

5.2 Cleaning and drying. Before the application of any preservative, all surfaces to which a preservative is to be applied shall be cleaned and dried in conformance with MIL-P-116, using cleaning method C-1 and any applicable drying method listed therein.

5.3 Preservation and packaging. Levels shall be as specified (see 6.1).

5.3.1 Level A.

5.3.1.1 Repainting. Damaged portions of painted surfaces shall be treated and painted in accordance with MIL-T-704, type A.

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5.3.1.2 Pump. All internal surfaces of the pump, including internal surfaces of the air chamber, shall be flushed with type P-10, type I, grade 30 preservative oil conforming to MIL-L-21260, actuating the pump during this operation to insure that all internal surfaces and operating mechanisms have been coated. All excess oil shall be drained. Threads of drain plugs shall be coated with the same type of oil and the plugs replaced and tightened. Openings into the interior of the pump shall be covered with metal or plastic caps or plugs. Prior to the threaded openings being covered, the threaded surfaces shall be coated with type P-2 preservative compound conforming to MIL-P-116. Nameplates shall be coated with a clear acrylic coating compound conforming to MIL-C-17504.

5.3.1.3 Unit packaging. Upon completion of pump preservation, the pumps shall be individually packaged within a fiberboard box conforming to RSC, type CF, class weather-resistant variety SW, grade 275 of PPP-B-636. Internal fiberboard blocking fabricated from material conforming to PPP-F-320 shall be used as required to insure a tight package. The boxes shall be closed in accordance with the appendix to PPP-B-636 for exterior boxes. In addition, all seams of the outer box, including the manufacturer's joint, shall be sealed in accordance with the appendix of PPP-B-636. The boxes shall not be strapped unless the boxes item is shipped without further protection (see 5.4.2).

5.3.2 Level C. The pump shall be cleaned and dried as specified in 5.1 and preserved in accordance with the manufacturer's commercial practice.

5.4 Packing. Levels shall be as specified (see 6.1).

5.4.1 Level A. Pumps, unit packaged as specified in 5.3, shall be packed not to exceed the gross weight limitations in snug-fitting, overseas-type containers conforming to PPP-B-585, PPP-B-591, PPP-B-601, or PPP-B-621. The boxes shall be closed and strapped in accordance with the applicable container specification and the appendix thereto, using galvanized strapping. The wood boxes or wood parts thereof shall be treated with water repellent wood preservative conforming to composition A of TT-W-572.

5.4.2 Level B. Pumps, unit packaged as specified in 5.3.1, may be shipped as individual units without further packing. When it is desired to ship two or more pumps in one shipping container, the pumps shall be packed as specified in 5.4.1 except that domestic-type containers shall be used, but water-repellent wood preservative treatment is not required.

5.4.3 Level C. Pumps, packaged as specified in 5.3, shall be packed to insure safe arrival at first destination for immediate use. Containers shall comply with Uniform Freight Classification rules or regulations of other carriers applicable to the mode of transportation.

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5.5 Marking. In addition to any special marking required by the contract or order, unit containers shall be marked in accordance with MIL-STD-129. Shipping containers shall be marked in accordance with Drawing D150-2-1.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Level of preservation, packaging, and packing required.
(c) Preproduction.

(1) Time allowed for suppliers to submit samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instruction when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether to proceed with regular production.

6.2 International interest. Certain provisions of this specification are the subject of ABC international standardization agreement. When amendment, revision, or cancellation of this specification is proposed, the departmental custodians will inform their respective Departmental Standardization Offices so that appropriate action may be taken respecting the international agreement concerned.

Custodian:

Army - MU
Air Force - 84

Preparing activity:

Army - MU(EA)

Review activity

Air Force - 84

User activities

Army - SM
Navy - MC

Project No. 1040-0082

International interest (see section 6)

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Edgewood Arsenal
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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of *delay and at the least cost.* Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?
   ☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 54 WHICH MAY BE USED.

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